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Efficient methanolysis of PET to produce high-purity DMT *via* homo-heterogeneous Mn^{2+}/Mn_2O_3 synergistic catalysis

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The depolymerization of waste polyethylene terephthalate (PET) has predominantly focused on zinc-based catalysts, benefiting from their stable Lewis acidity derived from unique 3d orbital electronic configurations. In contrast, manganese-based catalytic materials, which share similar electronic structural features, have rarely been explored. This study developed a synergistic catalytic system using $MnCl_2{\cdot}4H_2O$ and heterogeneous Mn_2O_3 for the methanolysis of PET which achieved complete PET conversion within 1 hour at 180 °C with a dimethyl terephthalate (DMT) yield of 95.35%. Compared with other catalytic systems for PET depolymerization, the present system uniquely integrates the low-cost and commercially available homogeneous Mn^{2+} ions and heterogeneous Mn_2O_3 oxide, combining the capability of rapid methanol activation and PET swelling with the structural stability and ease of recovery inherent to the heterogeneous component. Through systematic *in situ* and *ex situ* characterization studies (XRD, XPS, *in situ* DRIFTS, *etc.*), it was revealed that the essence of the synergistic effect originates from a dynamic composite active center formed at the interface between Mn^{2+} and Mn_2O_3, which efficiently stabilizes key reaction intermediates and promotes product desorption. The Aspen Plus process simulation and economic analysis further validate its industrial potential: despite the introduction of an additional step for homogeneous catalyst recovery, the synergistic system achieves superior product yield and process efficiency, translating into compelling economic advantages and strong commercial viability. In summary, this work establishes a novel homo-heterogeneous catalytic system, offering a strategic approach to develop efficient and scalable solutions for polyester degradation.

Green foundation

1. We have developed a synergistic catalytic system using a low-cost, commercially available manganese salt ($MnCl_2{\cdot}4H_2O$) and manganese oxide (Mn_2O_3) for the methanolysis of PET. This system achieves complete PET conversion with a 95.4% dimethyl terephthalate (DMT) yield at 180 °C within 1 h, enabling an economically viable and green pathway for polyester degradation without the need for complex synthesis or external reducing agents.
2. This study elucidates the unique "Mn^{2+}–O–Mn^{3+}" interfacial synergistic mechanism, wherein the homogeneous Mn^{2+} component interacts electronically with the heterogeneous Mn_2O_3 surface, inducing additional oxygen vacancies and facilitating rapid PET swelling. This mechanism enables efficient activation and conversion through interfacial synergy rather than external energy input, offering a new design concept that can be extended to a variety of catalytic materials and paving the way for the development of low-cost, stable catalysts for polyester degradation.
3. Aspen Plus simulation and economic analysis confirm that this process enables closed-loop recycling of materials and the production of high-purity DMT (99.99%). The use of a water/cyclohexane biphasic system not only allows complete solvent recycling with minimal loss but also effectively removes chloride ions, reducing corrosion risks and enhancing overall greenness. Despite the introduction of limited separation units, the process demonstrates clear economic competitiveness, providing a practical and sustainable solution for industrial-scale closed-loop plastic recycling.

Introduction

Polyethylene terephthalate (PET), one of the world's highest-production synthetic polyesters, is widely used in the packaging, textile, and pharmaceutical industries due to its excellent physical and chemical properties. According to the report of the Global Environmental Protection Research Network, the

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market size of the PET recycling industry has exceeded 80 billion US dollars in 2025. However, PET's high chemical stability also makes it difficult to degrade in the natural environment, with about 1800×10^4 tons of PET discarded every year.[1] The resulting "white pollution" and "microplastics" have become serious global environmental problems, and it is estimated that the global waste plastic will exceed 10 billion tons by 2050.[2,3] In response to these problems, various methods for PET recovery have been developed, such as mechanical recovery,[4] energy recovery, biological recovery[5–8] and chemical recovery.[9–12] Among these recycling strategies, chemical recovery boasts distinct advantages including relaxed feedstock prerequisites, rapid degradation kinetics, and high atom utilization efficiency. It can directly produce the monomeric constituents of polyethylene terephthalate (PET), thereby enabling both closed-loop recycling and value-added upcycling of this polymer. Accordingly, this approach has garnered extensive research interest over the past few years. Based on the degradation systems employed, current research on the chemical recycling of PET can be categorized into hydrolysis,[9] methanolysis,[13,14] glycolysis,[15–17] ammonolysis (or aminolysis)[18] and hydrogenolysis,[19,20] with a wide range of catalysts being utilized in these processes.

Among various chemical recycling routes, methanolysis has garnered significant attention due to its ability to directly produce dimethyl terephthalate (DMT), which is easy to purify and repolymerize. The rate-determining steps of this process typically involve the swelling of PET and the polarization/activation of its ester carbonyl group (C═O) by catalysts. Currently, high-performance catalysts are predominantly concentrated in zinc-based (*e.g.*, $Zn(OAc)_2$,[21,22] $ZnCl_2$,[23] ZnO,[18,24,25] Zn-based MOFs,[26] and Zn-based polyoxometalates[27–29]) and titanium-based systems (*e.g.*, organotitanium[30–33] and titanosilicates[34,35]). Their high performance originates from the strong and stable Lewis acidity provided by the central metal ions (Zn^{2+}, $3d^{10}$; Ti^{4+}, $3d^0$), which effectively lowers the electron cloud density of the carbonyl carbon in PET, facilitating nucleophilic attack by methanol. Generally, for the catalysts mentioned above, soluble metal ions can directly substitute the hydroxyl hydrogen of the alcohol to form an M–O–R structure, enhancing the electron density on the oxygen atom, which makes methanol more prone to nucleophilically attack the carbonyl carbon of PET, disrupting its long-chain structure, penetrating the PET skeleton, and causing rapid swelling. In contrast, heterogeneous materials must undergo mass transfer steps like solvent adsorption to exert their effect of activating the alcoholic hydroxyl group, which greatly limits their activity in depolymerizing PET. It is noteworthy that manganese ions (particularly Mn^{2+} with its $3d^5$ electronic structure) share a high similarity with Zn^{2+} in ionic radius and coordination geometry, theoretically possessing comparable potential for coordination activation. However, in PET depolymerization, manganese-based catalysts have long underperformed. Sean Najmi *et al.*[36] achieved nearly 100% selectivity for the BHET monomer only by using manganese oxide (MnO) under microwave-assisted heating for glycolysis, attributing the result to the rate-determining step being limited to the swelling stage, requiring auxiliary means to overcome mass transfer limitations. Daniel L. Lourenço *et al.*[37] demonstrated the catalytic activity of various commercial homogeneous and heterogeneous manganese materials towards various polyesters, but for PET specifically, its low solubility significantly impacts the reaction efficiency of single-component manganese-based systems. In addition, they have found that with potassium *t*-butoxide (KO^tBu), homogeneous manganese-based catalysts, including $Mn(OTf)_2$, $Mn_2(CO)_{10}$, $MnBr(CO)_5$ and $MeCpMn(CO)_3$, as well as the heterogeneous Si–Mn catalyst, can efficiently promote the depolymerization of polyesters. The heterogeneous catalyst maintains stable performance without obvious attenuation after 12 recycling cycles. Although KO^tBu was adopted as an additive in that work, this finding fully demonstrates the great potential of manganese-based catalysts for polyester degradation.[38]

In our study, we similarly found that single homogeneous or heterogeneous manganese-based materials exhibit low activity for PET methanolysis. However, adding a small amount of homogeneous manganese ions (Mn^{2+}) to a heterogeneous Mn oxide system substantially enhances both the PET depolymerization rate and DMT monomer yield. Such a homogeneous–heterogeneous synergistic system offers a promising strategy to overcome the inherent limitations of single-component catalysis. By combining the rapid molecular activation capability of soluble metal ions with the structural stability and easy recoverability of a solid support, this approach enables efficient reaction initiation while simplifying catalyst separation and reuse. Moreover, when the heterogeneous component is sufficiently stable, this design can avoid the unpredictability associated with catalysts that rely on uncontrolled metal leaching, thereby offering the potential for more consistent reaction kinetics and product quality.

Building on this rationale, this study proposes a synergistic catalytic system based on a homogeneous manganese salt ($MnCl_2 \cdot 4H_2O$) and heterogeneous manganese oxide (MnO_x) for PET degradation in methanol, which is a solvent with enormous industrial application potential.[12,39,40] We systematically designed control experiments (including comparisons of individual and combined catalytic components) to elucidate the synergistic mechanism between the homogeneous and heterogeneous components. By employing multiple characterization techniques (*e.g.*, XRD, XPS, TEM, and *in situ* FT-IR), we monitored the structural evolution of the catalysts and the variation in product distribution during the reaction, confirmed the existence of synergistic effects and demonstrated that the synergistic catalysis process does not lead to the destruction of the product and catalyst structures. Meanwhile, we have also confirmed that this synergistic system exhibits extremely high activity towards commercial plastics, and no significant decrease in its catalytic activity was observed over five consecutive reaction cycles. Furthermore, we conducted an industrial simulation using Aspen Plus to model the entire degradation process. The economic comparison between the synergistic and the standalone heterogeneous system revealed a clear advantage for the former. This advantage stems not only from the significantly enhanced PET degradation efficiency, which requires only one additional separation unit, but also from the inherently low cost and commercial availability of both $MnCl_2 \cdot 4H_2O$ and Mn_2O_3 catalysts. The introduction of this homogeneous catalyst recovery step is compatible with existing process flows, presenting considerable convenience and strong potential for future industrial implementation.

Experimental section

Materials

PET pellets were obtained from Titan Scientific Co., Ltd (Shanghai, China) with an intrinsic viscosity of 0.82 dL g^{-1} and analytical purity. Commercial plastic materials, primarily consisting of readily available mineral water bottles and various beverage bottles, were cut into pieces and collected for the reaction. Methanol (CH_3OH, AR) was purchased from Kelong Chemical Co., Ltd (Chengdu, China). Manganese(II) oxide (MnO) was purchased from Titan Scientific Co., Ltd (Shanghai, China). Manganese(II,III) oxide (Mn_3O_4, >99%) was supplied by Zancheng (Tianjin) Technology Co., Ltd. Manganese(III) oxide (Mn_2O_3, >98%) was obtained from Dingsheng Times Technology Co., Ltd (Chengdu, China). Manganese(IV) oxide (MnO_2, >98%) was obtained from Kelong Chemical Co., Ltd (Chengdu, China). Manganese(II) chloride tetrahydrate ($MnCl_2 \cdot 4H_2O$, AR) was purchased from Zancheng (Tianjin) Technology Co., Ltd. Manganese(II) oxalate dihydrate ($Mn(CH_3COO)_2 \cdot 2H_2O$ > 99%) was purchased from Macklin Biochemical Technology Co., Ltd (Shanghai, China). Manganese(II) nitrate tetrahydrate ($Mn(NO_3)_2 \cdot 4H_2O$ > 98%) was supplied by Aladdin Biochemical Technology Co., Ltd (Shanghai, China). Manganese(II) sulfate dihydrate ($MnSO_4 \cdot 2H_2O$ > 98%) was sourced from Sinopharm Chemical Reagent Co., Ltd (Shanghai, China). Chloroform ($CHCl_3$ > 99.5%) was obtained from Kelong Chemical Co., Ltd (Chengdu, China). High-purity argon gas (Ar > 99.999%) was provided by Changhong Gas Co., Ltd (Chengdu, China). Iron (III) oxide (Fe_2O_3, 99%) was purchased from Kelong Chemical Co., Ltd (Chengdu, China). Cobalt(III) oxide (Co_2O_3, 99%) was also obtained from Kelong Chemical Co., Ltd (Chengdu, China). Nickel(III) oxide (Ni_2O_3, LR) was supplied by Macklin Biochemical Technology Co., Ltd (Shanghai, China). Iron(III) chloride hexahydrate ($FeCl_3 \cdot 6H_2O$, AR, 99.0%) was sourced from General-Reagent Chemical Co., Ltd (Shanghai, China). Cobalt(II) chloride hexahydrate ($CoCl_2 \cdot 6H_2O$, 99%) was purchased from Kelong Chemical Co., Ltd (Chengdu, China). Nickel(II) chloride hexahydrate ($NiCl_2 \cdot 6H_2O$, 99%) was obtained from Adams Chemical Co., Ltd (Shanghai, China).

Methanolysis procedure

The standard procedure for catalytic methanolysis of PET is as follows: 1.0 g of PET pellets and 40 mL of methanol were added into a 100 mL high-pressure autoclave liner. A catalyst loading amounting to 10 wt% of PET was introduced, consisting of a mixture of homogeneous $MnCl_2 \cdot 4H_2O$ catalyst (1/10 of the total catalyst mass) and heterogeneous Mn_2O_3 catalyst (9/10 of the total catalyst mass), with a magnetic stirring bar placed inside the liner. After assembling the autoclave, the internal air was purged using argon gas for 15 minutes. Then the reaction mixture was heated to a target temperature (160 °C, 170 °C, 180 °C, 190 °C) for a specified holding time (0.5 h, 1 h, 2 h, 3 h, or 4 h). Upon reaching the target temperature, stirring was initiated at 600 rpm. After the reaction, the autoclave was allowed to cool to room temperature, and the internal pressure was confirmed to drop below 0.03 MPa. The reaction mixture was collected with a certain amount of chloroform added and then stirred for 30 minutes to ensure the complete dissolution of the products. Subsequently, the heterogeneous catalyst was first recovered through centrifugation and washing; the resulting liquid phase, part of which was subjected to chromatographic analysis for product distribution determination, had its residual portion dried at 70 °C overnight to afford the product crystals.

The PET conversion was determined by measuring the mass difference of the solids before and after the reaction. The main product dimethyl terephthalate (DMT) and byproducts mono(2-hydroxyethyl) terephthalate (MHET) and bis(2-hydroxyethyl) terephthalate (BHET) were quantified using gas chromatography (GC) with 1,2,3,4-tetrahydronaphthalene (THN) as the internal standard. The normalization method was applied with correction factors derived from the internal standard measurements. To further validate the accuracy, the product yields were also determined independently *via* the internal standard method. For ethylene glycol (EG), the yield was determined exclusively using the internal standard method with GC, and the results showed good consistency with those obtained from the calibrated normalization approach. The corresponding formulas are given below:

$$X_{\mathrm{PET}} = \frac{W_{\mathrm{PET}} + W_{\mathrm{CAT}} - W_{\mathrm{S}}}{W_{\mathrm{PET}}} \times 100\% \tag{1}$$

$$Y_{\mathrm{DMT}} = \frac{W_{\mathrm{DMT}}}{W_{\mathrm{DMT}} + W_{\mathrm{MHET}} + W_{\mathrm{BHET}}} \times X_{\mathrm{PET}} \tag{2}$$

$$Y'_{\mathrm{DMT}} = \frac{W_{\mathrm{DMT}}}{W'_{\mathrm{DMT}}} \tag{3}$$

$$Y'_{\mathrm{EG}} = \frac{W_{\mathrm{EG}}}{W'_{\mathrm{EG}}} \tag{4}$$

where X_{PET} is the conversion rate of PET, W_{PET} represents the mass of PET before the reaction, W_{CAT} refers to the mass of the heterogeneous catalyst before the reaction, W_{S} stands for the mass of the solid recovered after the reaction (including unreacted PET and catalyst), Y_{DMT} refers to the DMT yield calculated using the normalization method, W_{DMT} and W_{EG} are the masses of DMT and EG after the reaction, Y'_{DMT} and Y'_{EG} indicate the DMT and EG yields determined using the internal standard method, and W'_{DMT} and W'_{EG} denote the theoretical masses of DMT and EG producible. The yields of all other byproducts were calculated following the same procedure.

Meanwhile, we performed first-order kinetic fitting and calculated the corresponding reaction rate constant and activation energy *via* the Arrhenius equation, as shown in the equations below:

$$\ln\left(\frac{1}{1-X_{\mathrm{PET}}}\right)=kt \tag{5}$$

$$\ln k=\ln A-\frac{E_{\mathrm{a}}}{RT} \tag{6}$$

where k, A, E_a, T and R represent the reaction rate constant, pre-exponential factor, apparent activation energy, absolute temperature (in Kelvin) and the universal gas constant, respectively.

Catalyst recycling procedure

Recovery of the catalyst. The recycling of the catalytic system involved separate treatments for its heterogeneous and homogeneous components. The heterogeneous Mn_2O_3 was readily recovered by simple centrifugation, washing, and drying, allowing for straightforward reuse. For the homogeneous $MnCl_2·4H_2O$, direct recovery as a solid was challenging due to its trace amount and the tendency to form a mixture with DMT crystals upon product drying. Instead, repeated water washing was employed to remove $MnCl_2$ from the product, and inductively coupled plasma optical emission spectrometry (ICP-OES) was used to determine the residual manganese content in the washed product.

Cycling stability test. To evaluate the long-term stability of the catalytic system, a series of consecutive experiments were conducted with partial replenishment of both components. After each reaction, the mixture was cooled for 30 minutes and centrifuged to separate the solid products. For the homogeneous component, 20 mL of the supernatant methanol solution was collected and mixed with an equal volume (20 mL) of fresh methanol solution containing the same concentration of Mn^{2+} ions, yielding a total of 40 mL for the subsequent reaction. For the heterogeneous component, the recovered solids were dissolved in chloroform to remove organic products, followed by centrifugation and drying. Half of the recovered Mn_2O_3 was then combined with an equal amount of fresh Mn_2O_3 for the next cycle. This protocol simulated the partial catalyst replenishment expected in continuous processes and effectively validated the cycling stability of the system.

To further rigorously illustrate the stable activity of the catalyst, we designed two additional control experiments: (1) using recovered Mn_2O_3 and fresh $MnCl_2·4H_2O$, and (2) using recovered Mn_2O_3 and recovered $MnCl_2·4H_2O$. Since each recycling experiment inevitably causes catalyst loss, the feeding amounts of reactants and catalysts were reduced proportionally for each group, and three recycling runs were performed in each test.

Aspen Plus process simulation. To further evaluate the potential industrial application value of the synergistic system and propose rational improvements for real-world industrial processes, this study employed Aspen Plus V14 for process simulation based on experimental data.

Analytical methods

A range of analytical techniques were used in this research. The composition of the methanolysis product of PET was determined by gas chromatography (GC2060). The crystal structures of the catalysts and degraded products were analyzed by X-ray diffraction (XRD, Rigaku MiniFlex 600, Cu-Kα radiation, λ = 0.154 nm). Surface morphologies of the catalysts and partially reacted PET were examined by field-emission scanning electron microscopy (FE-SEM, Hitachi Regulus 8230). The specific surface area of the catalysts was determined using high-performance specific surface area and micropore analyzer (BET, BSD-660M A3M analyzer, N_2 adsorption). Inductively coupled plasma optical emission spectrometry (ICP-OES, PerkinElmer Avio 200) was used to quantify manganese leaching. Chemical state changes of the catalysts before and after the reaction were investigated by X-ray photoelectron spectroscopy (XPS). Surface oxygen defects on catalysts were obtained through electron paramagnetic resonance (EPR, EMX Plus). Acidic properties, including acid type and distribution, were probed using pyridine-adsorbed FTIR spectroscopy (Bruker Tensor 27/Vertex 80v). The acidity and alkalinity were obtained using a high-performance chemical adsorption instrument (BelCata II, Japan, 50–800 °C). Molecular weight changes of PET at different reaction times were monitored by gel permeation chromatography (GPC, Agilent PL-GPC 50) using hexafluoroisopropanol as the mobile phase. The functional groups of the reaction products were identified by Fourier transform infrared spectroscopy (FT-IR, PerkinElmer Spectrum 3). The product structure was obtained by testing with a nuclear magnetic resonance spectrometer (400M NMR, JNM-ECZ400S/L). Furthermore, *in situ* diffuse reflectance infrared Fourier transform spectroscopy (*in situ* DRIFTS) was conducted to monitor real-time surface changes (spectral range: 600–4000 cm^{-1}, resolution: 4 cm^{-1} 64 scans, under argon purge).

Results and discussion

Synergistic degradation of PET by MnO_x and Mn^{2+}

Fig. 1 illustrates the synergistic catalytic performance of manganese oxides (MnO_x) with different valence states and $MnCl_2·4H_2O$ in PET degradation. As shown in Fig. S1, the liquid products were analyzed by gas chromatography. The main product, dimethyl terephthalate (DMT), eluted at 9–10 min, while the byproducts, ethylene glycol (EG), mono(2-hydroxyethyl) terephthalate (MHET) and bis(2-hydroxyethyl) terephthalate (BHET), eluted at 3.4 min, 12.4 min and 14.1 min, respectively.

The synergy between MnO_x and $MnCl_2·4H_2O$ was first evaluated at 170 °C (Fig. 1a). When MnO_x alone was used (10 wt% of PET), the PET depolymerization degree for all oxides was below 20%. However, upon the addition of a small amount of

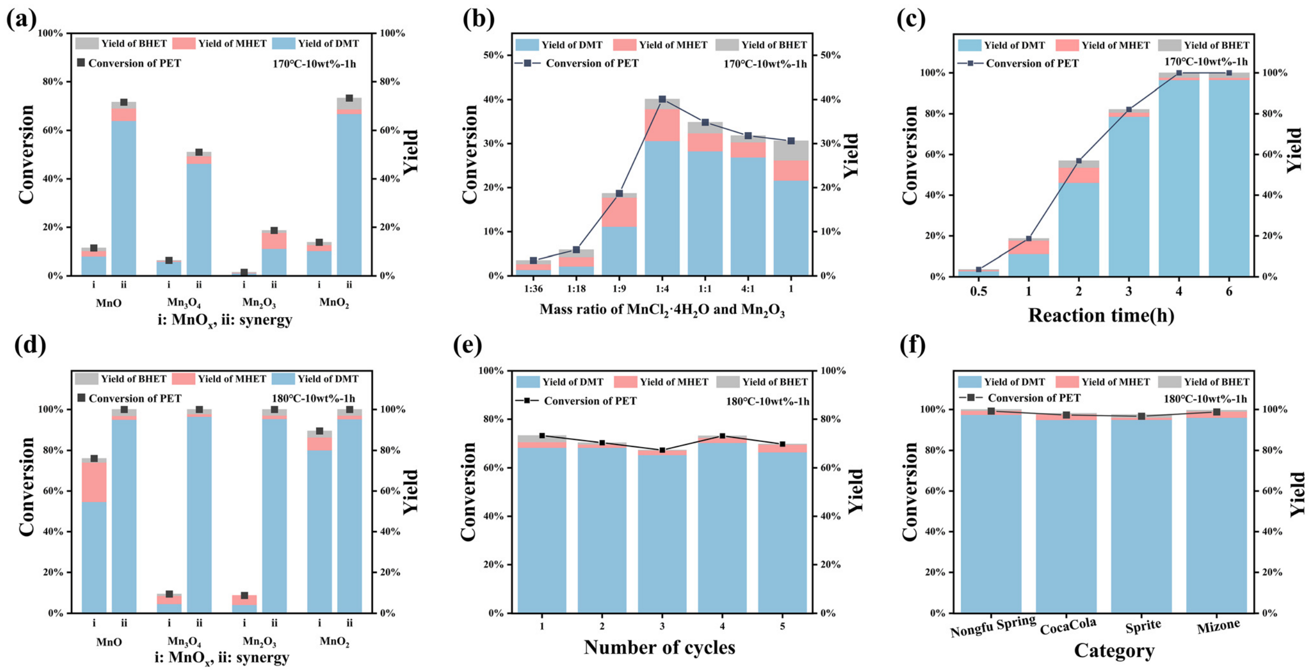

Fig. 1 Comparison of the heterogeneous system and the synergistic system at (a) 170 °C and (d) 180 °C (where 'i' represents the individual MnO_x catalyst and 'ii' represents the synergistic catalyst); (b) mass ratio of $MnCl_2·4H_2O$ and Mn_2O_3 at 170 °C; (c) reaction time of the synergistic system at 170 °C; (e) performance of the synergistic system over repeated catalyst recycling at 180 °C for 1 hour; (f) degradation performance of commercial plastics at 180 °C for 1 hour.

$MnCl_2·4H_2O$ (constituting 1/10 of the total catalyst mass), the depolymerization degree increased significantly. Notably, Mn_2O_3 exhibited the most dramatic change: its depolymerization degree was less than 1% when used alone, but increased to 18.68% in synergy with Mn^{2+}, which was significantly higher than that achieved by using an equivalent amount of $MnCl_2·4H_2O$ alone (13.96%). This provides initial evidence for a synergistic effect. At an elevated temperature of 180 °C, the synergy became more pronounced (Fig. 1d), as the PET conversion rate increased from 8.72% to 100% and reached a DMT yield of 95.35%. The reaction conditions were superior to many reported homogeneous or even homogeneous catalytic systems, as shown in Table 1.

Other manganese oxides (*e.g.*, MnO and MnO_2) also exhibited some degree of synergy. Even their leaching may exert a synergistic effect with their heterogeneous fraction, thus resulting in a relatively high conversion rate when they exist alone (Fig. 1d), but their high solubility in the high-temperature methanol (Tables 2 and 3, [Mn^{2+}] in theory was calculated using concentrations of different solutions) gives rise to uncontrollability in continuous industrial production, which may lead to fluctuations in reaction kinetics, frequent catalyst

Table 1 Comparison of catalytic performance for PET alcoholysis with the reported systems

Types of catalysts	Temperature (°C)	Time (min)	X_{PET} (%)	$Y_{monomer}$ (%)
Mn_2O_3 & $MnCl_2·4H_2O$ (this work)	180	60	100	95[a]
Cu/ZnO[41]	220	360	97	74[a]
Cu/SiO_2 [42]	200	90	92	91[a]
Fe_2O_3@MnS_2 [43]	225	180	97	90[b]
MgO/NaY[44]	200	30	99	91[a]
OPA@Fe_3O_4 [45]	200	60	100	83[a]
Engineered hydroxyapatite (HAP)[46]	180	120	100	90[a]
$K_{10}[Zn_4(H_2O)_2(PW_9O_{34})_2]·H_2O$[47]	240	480	100	93[b]
$Na_{12}[WZn_3(H_2O)_2(ZnW_9O_{34})_2]$[29]	190	40	100	85[b]
Protic ionic liquids (PIL3)[48]	190	30	100	87[b]
Non-stoichiometric protic ionic liquids[49]	180	180	100	97[a]
Pro/Zn$(Ac)_2$[50]	210	15	100	67[b]
$Zn(OTf)_2$ & GVL[51]	180	300	100	82[c]
70% $ZnCl_2/H_2O$[23]	180	480	98	97[c]

The subscripts 'a', 'b', and 'c' represent the products methanol, ethylene glycol and terephthalic acid, respectively.

Table 2 Dissolution rate of Mn^{2+} when manganese oxide is used alone

MnO_x	$[Mn^{2+}]$ after reaction (mg L^{-1})
MnO	70.52
Mn_3O_4	5.37
Mn_2O_3	2.31
MnO_2	41.51

Reaction conditions: 180 °C, 1 h.

Table 3 Dissolution rate of Mn^{2+} when $MnCl_2{\cdot}4H_2O$ is added

Catalyst	Mass of $MnCl_2{\cdot}4H_2O$ (g)	$[Mn^{2+}]$ in theory (mg L^{-1})	$[Mn^{2+}]$ in reality (mg L^{-1})
$MnCl_2{\cdot}4H_2O$	0.0102	56.54	56.29
MnO	0.0106	62.19	91.60
Mn_3O_4	0.0103	52.89	64.86
Mn_2O_3	0.0112	54.44	55.26
MnO_2	0.0109	59.77	78.63

First row: 1 wt% $MnCl_2{\cdot}4H_2O$ alone (180 °C, 1 h); all others: synergistic reactions (180 °C, 1 h).

replenishment, and even adverse effects on the product structure when the leaching concentration exceeds the threshold. In contrast, Mn_2O_3 exhibited minimal leaching and excellent structural stability under the reaction conditions (even with $MnCl_2{\cdot}4H_2O$ added), making it more suitable for industrial applications requiring catalyst stability. Therefore, the Mn_2O_3/$MnCl_2{\cdot}4H_2O$ system was selected for further optimization.

Subsequently, the mass ratio of $MnCl_2{\cdot}4H_2O$ to Mn_2O_3 was optimized (Fig. 1b). At 170 °C, a 1 : 4 ratio gave the highest PET degradation degree (40.11%). However, further ^{1}H NMR analysis of the products revealed significant broadening of the characteristic DMT signals (aromatic and methoxy protons) at this ratio (Fig. S15). This broadening is attributed to the paramagnetic relaxation enhancement effect of residual Mn^{2+} ions, which shorten the relaxation times of nearby ^{1}H nuclei, leading to peak broadening and a reduced signal-to-noise ratio. When the $MnCl_2{\cdot}4H_2O$ to Mn_2O_3 ratio reached 1 : 1, the solvent peak ($CDCl_3$, δ = 7.26 ppm) exhibited distinct splitting, indicating that the concentration of free Mn^{2+} in solution had reached a critical threshold, where it interacts with the solvent *via* weak coordination or chemical exchange, thereby perturbing the local magnetic environment. A similar situation also appears in the ^{1}H spectrum of the product at 180 °C. These NMR observations collectively demonstrate that excessive homogeneous Mn^{2+} loading compromises product purity and structural integrity. Furthermore, to further verify the potential influence of homogeneous ions on product purity, we performed ICP tests to determine the residual manganese content in products under the two different ratios, and the results are summarized in Table S12. When the $Mn_2O_3 : MnCl_2{\cdot}4H_2O$ ratio was 1 : 9, three water washes dramatically reduced the Mn content from 2790 mg kg^{-1} to 70.1 mg kg^{-1}. Further increasing the washing times brought about no obvious decline in manganese content, and the residual Mn concentration remained at 27.2 mg kg^{-1} after ten washing cycles. In contrast, for the ratio 1 : 4, a manganese residue of 189 mg kg^{-1} was still observed even after ten washes. These findings confirm that an excessive dosage of manganese in the synergistic system leads to abundant Mn residue in products, which inevitably deteriorates product quality. Meanwhile, it also indicates that excessive addition of heterogeneous components will impose a higher cost burden on downstream separation in practical industrial processes. Considering both catalytic activity and product quality, the 1 : 9 ratio was selected as the optimal balance, where the DMT product exhibited sharp, well-resolved signals with no detectable peak broadening or solvent peak anomalies. Considering that excessively high chloride concentrations may cause corrosion in equipment in practical industrial applications, this study adopted a lower chloride concentration as the operating condition in the process simulation. Although previous studies have shown that low chloride concentrations (<2000 mg L^{-1}) have a limited corrosive effect on 316L stainless steel,[52] and the actual chloride concentration in this process (*ca.* 55.26 mg L^{-1}) was well below this threshold, the use of an even lower chloride concentration helps to further ensure long-term equipment stability, especially given that elevated temperatures can accelerate chloride-induced corrosion.[53]

To evaluate the cycling stability of the catalytic system, we first conducted five consecutive recycling experiments using a "semi-supplementation" strategy, with the results shown in Fig. 1e. Throughout the recycling tests, the PET conversion and DMT yield remained at approximately 70%, with no significant decline observed as the number of cycles increased. Compared with the first run, the conversion decreased by about 30%, which is mainly attributed to the inevitable presence of residual products such as DMT in the Mn^{2+}-containing methanol solution after the reaction, leading to a slight reduction in the effective concentration of the active species. Nevertheless, both PET conversion and DMT yield remained stable over subsequent cycles, demonstrating the good cycling stability of this catalytic system.

The rigorous catalyst stability test results are listed in Table S6. When the recovered Mn_2O_3 was combined with fresh $MnCl_2{\cdot}4H_2O$, PET conversion remained at approximately 97%, and the DMT yield was consistently higher than 91%, which was comparable to the performance of the initial reaction. This directly verifies the structural stability of recycled Mn_2O_3, with nearly no activity loss observed during the recycling process. When both the homogeneous and heterogeneous components were recycled, PET conversion and DMT yield show a slight downward trend with the increase in recycling cycles. This is mainly attributed to the unavoidable loss of Mn^{2+} during circulation, such as entrainment in products and solution loss in the post-treatment process, which further demonstrates the rationality of the "semi-supplementation" strategy.

In addition, the degradation performance of the synergistic system toward real commercial plastics was evaluated, as shown in Fig. 1f. Four types of common PET bottles were selected as substrates, and the PET conversion exceeded 97% in all cases, with DMT yields maintained at approximately 95%. These results indicate that the developed synergistic catalytic system exhibits excellent degradation performance toward real commercial plastics, highlighting its promising potential for industrial application.

For comparison, the performance of other homogeneous manganese salts ($MnSO_4{\cdot}2H_2O$, $Mn(NO_3)_2{\cdot}4H_2O$, and $Mn(CH_3COO)_2{\cdot}2H_2O$) was also evaluated (Fig. S8). These salts were inferior to $MnCl_2{\cdot}4H_2O$ in either activity or stability due to their low solubility in methanol ($MnSO_4{\cdot}2H_2O$), instability leading to product discoloration ($Mn(NO_3)_2{\cdot}4H_2O$), or side reactions with products ($Mn(CH_3COO)_2{\cdot}2H_2O$). Consequently, the subsequent industrial process analysis was based solely on the optimal $MnCl_2{\cdot}4H_2O/Mn_2O_3$ synergistic system. Furthermore, the general applicability of this synergistic concept was further explored. As shown in Fig. S9, three common transition metal oxides—iron(III) oxide (Fe_2O_3), cobalt(III) oxide (Co_2O_3), and nickel(III) oxide (Ni_2O_3)—were selected for comparative experiments on PET methanolysis, either alone or in combination with their corresponding chlorides ($FeCl_3{\cdot}6H_2O$, $CoCl_2{\cdot}6H_2O$, and $NiCl_2{\cdot}6H_2O$). In all three cases, the synergistic catalytic systems exhibited significantly enhanced performance compared to the single-component counterparts, with improved conversion and product yields. These results further validate the broad applicability of the homogeneous–heterogeneous synergistic strategy and highlight its potential for industrial application in polyester degradation.

Finally, the product yields for each reaction were rigorously quantified using the internal standard method, with the results presented in Fig. S3–5. Based on these data, the apparent activation energy of the synergistic catalytic system was determined to be 151.61 kJ mol^{-1} by fitting the kinetic data to a first-order model according to the Arrhenius equation. Compared to some reported low-temperature alcoholysis systems,[54–58] the reaction temperature and activation energy of this system are relatively high, but these low activation energy degradation systems often face significant solvent (*e.g.*, organic solvent or biological additives) or metal contamination. However, thanks to the commercial availability and easy separation characteristics of the catalyst, as well as the universal applicability of the synergistic system, our synergistic system still has great potential in industry.

Catalyst characterization

To elucidate the structural basis of the efficient PET methanolysis and the homogeneous–heterogeneous synergistic mechanism within the $Mn_2O_3/MnCl_2{\cdot}4H_2O$ system, the catalysts were systematically characterized before and after the reaction. The results demonstrate that Mn_2O_3 exhibits exceptional structural stability, suitable surface acidity, and a unique electronic structure, which are key to its high synergistic performance and industrial potential.

XRD analysis (Fig. 2a) revealed that the characteristic diffraction peaks of commercial α-Mn_2O_3 (PDF#00-041-1442) remained virtually unchanged after the reaction, both in standalone and synergistic systems, confirming its superior crystalline phase stability and corroborating the low leaching data. In contrast, other manganese oxides underwent significant transformations (Fig. S17): MnO_2 irreversibly transformed into the spinel Mn_3O_4 phase (Fig. S17c) while Mn_3O_4 itself exhibited increased crystallinity after the synergistic reaction (Fig. S17b), indicating that the catalyst underwent thermodynamic-driven structural optimization during the process,[59] which made the catalyst lose its rich defects and active sites associated with low crystallinity. MnO maintained the stability of its crystal structure before and after the reaction (Fig. S17a), but its high leaching rate limits its industrial application (Table 1). These comparisons underscore the unique advantage of Mn_2O_3 as a stable heterogeneous scaffold under harsh reaction conditions.

Pyridine-adsorbed IR spectroscopy (Py-IR, Fig. 2b) showed adsorption peaks at 1445, 1492, and 1605 cm^{-1}, indicating the co-presence of abundant Lewis (L) and Brønsted (B) acid sites on the Mn_2O_3 surface. In the methanolysis of PET, Brønsted (B) acid sites can protonate the ester carbonyl group (C═O) in PET molecules which increase the electrophilicity of the carbonyl carbon, making it more susceptible to nucleophilic attack by methanol, thereby accelerating ester bond cleavage. Meanwhile, Lewis (L) acid sites coordinate with and activate the oxygen atom of the ester group, which also enhances the electrophilicity of the carbonyl carbon and promotes nucleophilic attack by methanol. Additionally, L acid sites can adsorb and activate methanol molecules, further increasing the reaction rate.[60] Conventional FT-IR spectra (Fig. S18a) further supported the stability of Mn_2O_3: only a weak product adsorption peak at 1720 cm^{-1} appeared after the reaction, while the vibrations attributed to Mn^{3+}–O bonds (750–420 cm^{-1}) remained unchanged. Conversely, the surfaces of other manganese oxides were covered with substantial organic residues post-reaction (Fig. S18c), suggesting potential side reactions or difficult product desorption, which is detrimental to catalyst recycling.

X-ray photoelectron spectroscopy (XPS) provided electronic-scale insights into the synergy. The Mn 2p spectrum of Mn_2O_3 (Fig. 2c) shifted slightly toward lower binding energy after the reaction, while the Mn 3s multiplet splitting energy (ΔE) remained constant (Fig. 2d), indicating no change in the average manganese oxidation state. The O 1s spectrum (Fig. 2f) showed an increased peak at ∼531.7 eV after the reaction, especially under synergistic conditions, which is probably attributed to surface oxygen vacancies or defect oxygen species. Electron paramagnetic resonance (EPR) further confirmed the presence of oxygen vacancies (Fig. 2c). Fresh Mn_2O_3 exhibited a strong symmetric signal at $g \approx 2.003$, characteristic of trapped electrons at oxygen vacancies.[61] After the reaction, the EPR signal was enhanced in both the non-synergistic and synergistic systems, with the synergistic system showing a significantly larger increase, while the signal position remained unchanged. This indicates that the synergistic reaction induces additional oxygen vacancies without altering their

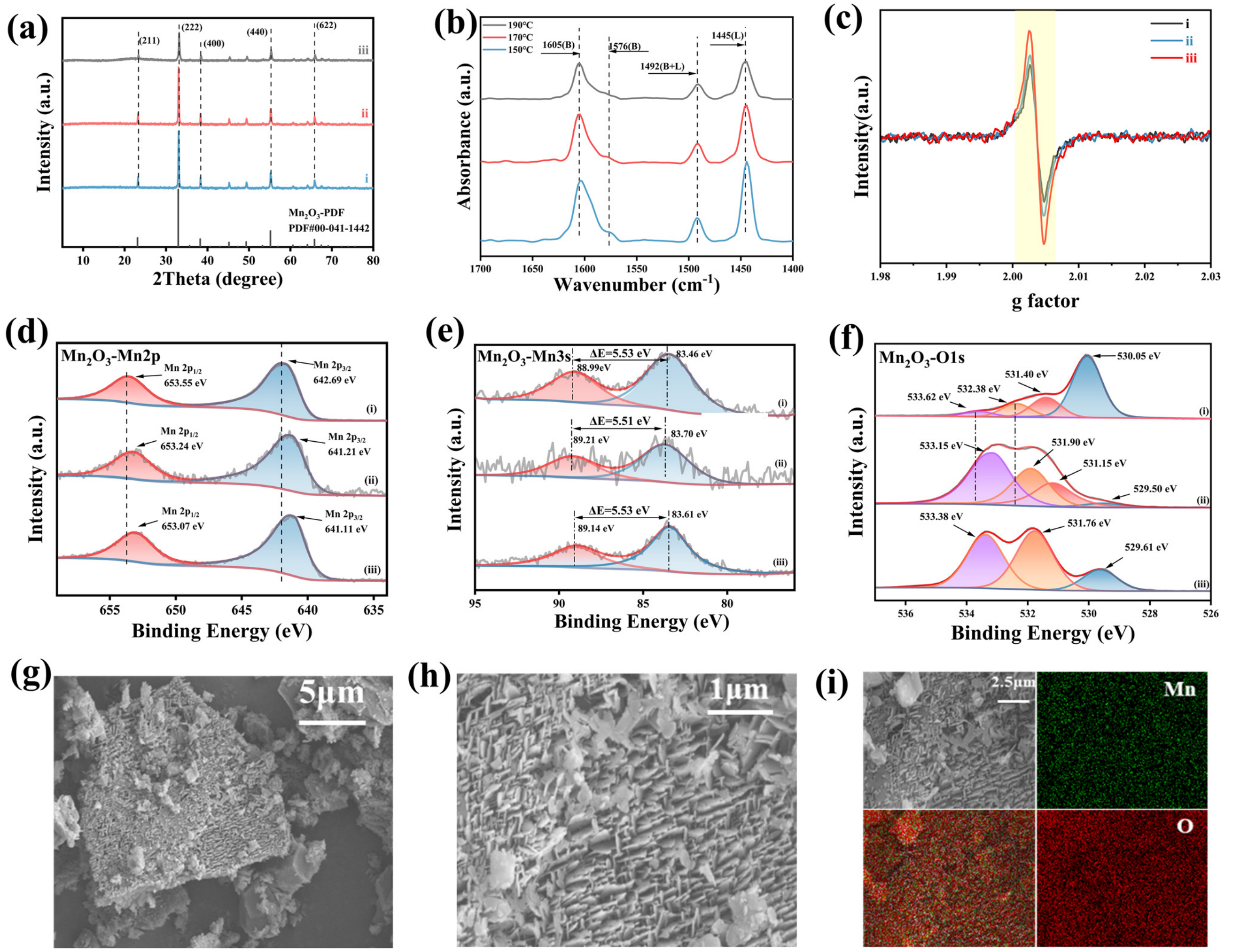

Fig. 2 (a) XRD patterns before and after the reaction; (b) pyridine infrared of Mn_2O_3 at 150 °C, 170 °C, and 190 °C; (c) EPR spectrum of Mn_2O_3; high-resolution XPS spectra of Mn_2O_3 for (d) Mn 2p, (e) Mn 3s and (f) O 1s ('i' denotes the catalyst before the reaction; 'ii' denotes the catalyst after the reaction in the single heterogeneous system; and 'iii' denotes the catalyst after the reaction in the synergistic system); (g–i) Mn_2O_3 SEM images and element distribution.

chemical nature. CO_2-TPD (Fig. S19b) further revealed a strong desorption peak at *ca.* 700 °C, consistent with oxygen-vacancy-induced basic sites. These oxygen vacancies modulate the local electronic structure, increasing electron density on neighboring Mn and O atoms (consistent with the lower XPS binding energy), thereby creating electron-rich active sites. Combined with Py-IR results, these sites can interact electronically with homogeneous Mn^{2+} at the reaction temperature (180 °C), promoting the formation of a dynamic "Mn^{2+}–O–Mn^{3+}" composite interface that efficiently activates PET ester bonds and stabilizes key intermediates, ultimately enhancing synergistic catalytic performance. In contrast, XPS analysis of MnO and MnO_2 (Fig. S21a and Fig. S21c) showed significant valence changes, indicating surface instability.

The SEM images (Fig. 2g–i) reveal the unique two-dimensional sheet-like morphology of Mn_2O_3, compared to the irregular particles of other oxides (Fig. S22). Its abundant edge defects correlate with the oxygen vacancies detected by XPS, collectively creating a micro-environment efficient for reactant activation. The regular 2D planes may facilitate the orderly adsorption and mass transfer of reactants and intermediates, achieving high catalytic efficiency while maintaining structural integrity. This advantage extends beyond the conventional pursuit of high surface area, offering favorable conditions for scale-up.[62] Furthermore, N_2 adsorption–desorption measurements were performed on all MnO_x catalysts, as shown in Fig. S23. The isotherm of Mn_2O_3 exhibited a typical type IV shape. Its narrow pore size distribution (centered at 5–20 nm) with the highest peak intensity confirmed a well-ordered mesoporous structure with a large pore volume, resulting from the stacking of its two-dimensional sheets, which is able to serve as the physical foundation for its role as a stable heterogeneous support.

In summary, the characterization results indicate that Mn_2O_3 in the synergistic system is not an inert support but an active heterogeneous component with structural stability, rich acid sites, and tunable oxygen vacancies. Its unique sheet-like morphology further optimizes the reaction interface. These properties, synergizing with homogeneous Mn^{2+}, collectively enable the efficient and highly selective depolymerization of PET, providing clear guidance for catalyst design in its industrial application.

Product characterization

To gain deeper insights into the degradation pathway of PET and the influence of the $Mn_2O_3/MnCl_2$ synergistic system on the product structure, the evolution of PET particles during the reaction and the final products were systematically characterized.

The morphological changes of PET particles during the reaction visually reflect the degradation mechanism. The SEM images (Fig. 3a) show that after 30 minutes of reaction, the PET particle surfaces exhibited clear signs of erosion while largely maintaining their bulk structure. When the reaction proceeded to 60–180 minutes, numerous cracks and fine fragments appeared on the particle surfaces, indicating that degradation had progressed inwards through the formation of a porous structure.[47] Consistent with this morphological evolution is the change in the weight-average molecular weight (M_w) of PET (Fig. 3b). During the initial stage (0–60 min), the decrease in M_w was slow, corresponding to the swelling and initial surface erosion of PET. Subsequently (60–180 min), M_w decreased rapidly in an approximately linear manner, indicating the massive cleavage of internal ester bonds. This kinetic behavior aligns well with a first-order reaction model, further supporting the kinetic analysis presented earlier.

The structure of the main product was confirmed by Fourier-transform infrared spectroscopy (FT-IR). Products from all synergistic systems showed strong absorption bands at 1720 cm^{-1} (ester C═O), 1280 cm^{-1} and 1120 cm^{-1} (C–O–C), and 1610, 1580, and 1500 cm^{-1} (aromatic ring vibrations); the characteristic doublet for *para*-disubstituted benzene at 870 and 730 cm^{-1} was particularly evident (Fig. 3c). These fingerprint peaks match perfectly with the standard spectrum of dimethyl terephthalate (DMT), proving that the primary product of PET methanolysis is high-purity DMT. Notably, this result corroborates the analysis of surface residues on the catalysts (Fig. S18): the Mn_2O_3 system showed complete product desorption, while other oxides (*e.g.*, MnO, Mn_3O_4, and MnO_2) exhibited significant organic residue peaks on their surfaces, again highlighting the potential advantage of the Mn_2O_3 system in terms of product separation. The XRD patterns of the products (Fig. 3d) show that as the reaction time increased to 4 hours, the diffraction peaks belonging to the monoclinic crystal planes of DMT (*e.g.*, (202), (040), and (060)) gradually intensified, indicating improved product crystallinity. However, extending the reaction beyond 4 hours led to a decrease in crystallinity and with no significant yield improvement (Fig. 3e). This suggests that 4 hours represents the optimal reaction time under the given conditions (170 °C, 10 wt%), achieving both a high DMT yield (96.53%) and favorable product crystallinity. Prolonged reaction times may induce side reactions or alter the crystal structure, offering no benefit to product quality. In addition, the products obtained from the cycling experiments and the degradation of commer-

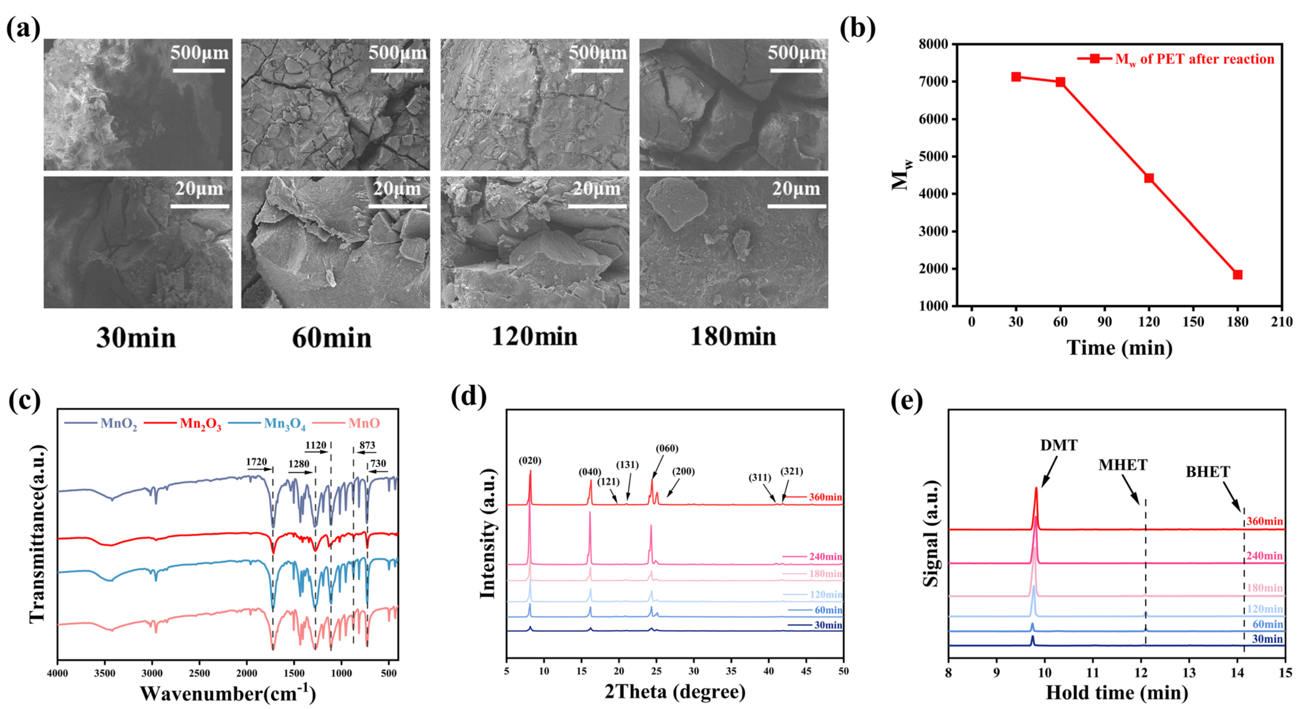

Fig. 3 (a) SEM image of PET and (b) weight-average molecular weight at different reaction times (170 °C, 10 wt% cat); (c) product chromatographic profile of different MnO_x in the synergistic reaction (170 °C, 10 wt% cat), (d) XRD patterns and (e) infrared spectra of the product at different reaction times (170 °C, 10 wt% cat).

cial plastics using the synergistic system were characterized by nuclear magnetic resonance (NMR) and X-ray diffraction (XRD), with the results shown in Fig. S24. Both products exhibited well-defined crystalline structures, which are in good agreement with the previously presented product yield data.

In situ infrared spectroscopy and mechanistic analysis

To further elucidate the species evolution and reaction mechanism during the synergistic process, and to distinguish it from the methanolysis catalyzed by MnO_x alone, *in situ* diffuse reflectance infrared Fourier transform spectroscopy (*in situ* DRIFTS) was employed for both systems. Integrating the catalytic performance and previously discussed characterization results, a reaction mechanism based on interfacial synergistic catalysis is proposed. Given the limited solubility of PET in methanol and the fact that methanolysis is essentially a transesterification reaction, bis(2-hydroxyethyl) terephthalate (BHET), which shares the core structural groups with PET, was selected as a model compound. A methanol solution of BHET (2 g L^{-1}) was carried into the *in situ* cell under argon flow, and the infrared signals of species on the catalyst surface were recorded during the temperature-programmed reaction, as shown in Fig. 4.

In the Mn_2O_3-only catalytic system (Fig. 4d), the reaction initiates with the competitive adsorption and preliminary activation of methanol and PET (or BHET) on the catalyst surface. The Lewis acid sites (Mn^{3+}) on the Mn_2O_3 surface can adsorb the carbonyl oxygen of the substrate, while its surface hydroxyl groups may interact with methanol. This adsorption configuration brings the oxygen atom of methanol close to the activated carbonyl carbon, enabling nucleophilic attack.[63] This leads to the gradual cleavage of the PET long chains, forming oligomers or BHET monomers, and subsequently generating the necessary tetrahedral transition state intermediate for transesterification, culminating in group exchange to yield the byproduct MHET and the main product DMT. However, the efficiency of this pathway is constrained by two factors: firstly, the methoxy species generated on the heterogeneous surface (Mn^{3+}–OCH_3) exhibit relatively weak nucleophilicity; secondly, the intermediates and products generated during the reaction interact strongly with the Mn_2O_3 surface, resulting in slow desorption. *In situ* spectroscopic data (Fig. 4b) show that the signal change around ~1649 cm^{-1}, attributed to the coordinated carbonyl group,[47] is minimal, confirming sluggish reactant conversion and intermediate evolution in this pathway. Similar trends were observed in the *in situ* DRIFTS of other MnO_x catalysts (Fig. S25–Fig. 27). In contrast, the $Mn_2O_3/MnCl_2{\cdot}4H_2O$ synergistic system exhibits a markedly different dynamic process. Firstly, during the reactant activation stage, the introduction of homogeneous Mn^{2+} does not replace the adsorption function of Mn_2O_3 but rather creates a crucial interfacial synergy with it. The Mn^{2+} ions in solution

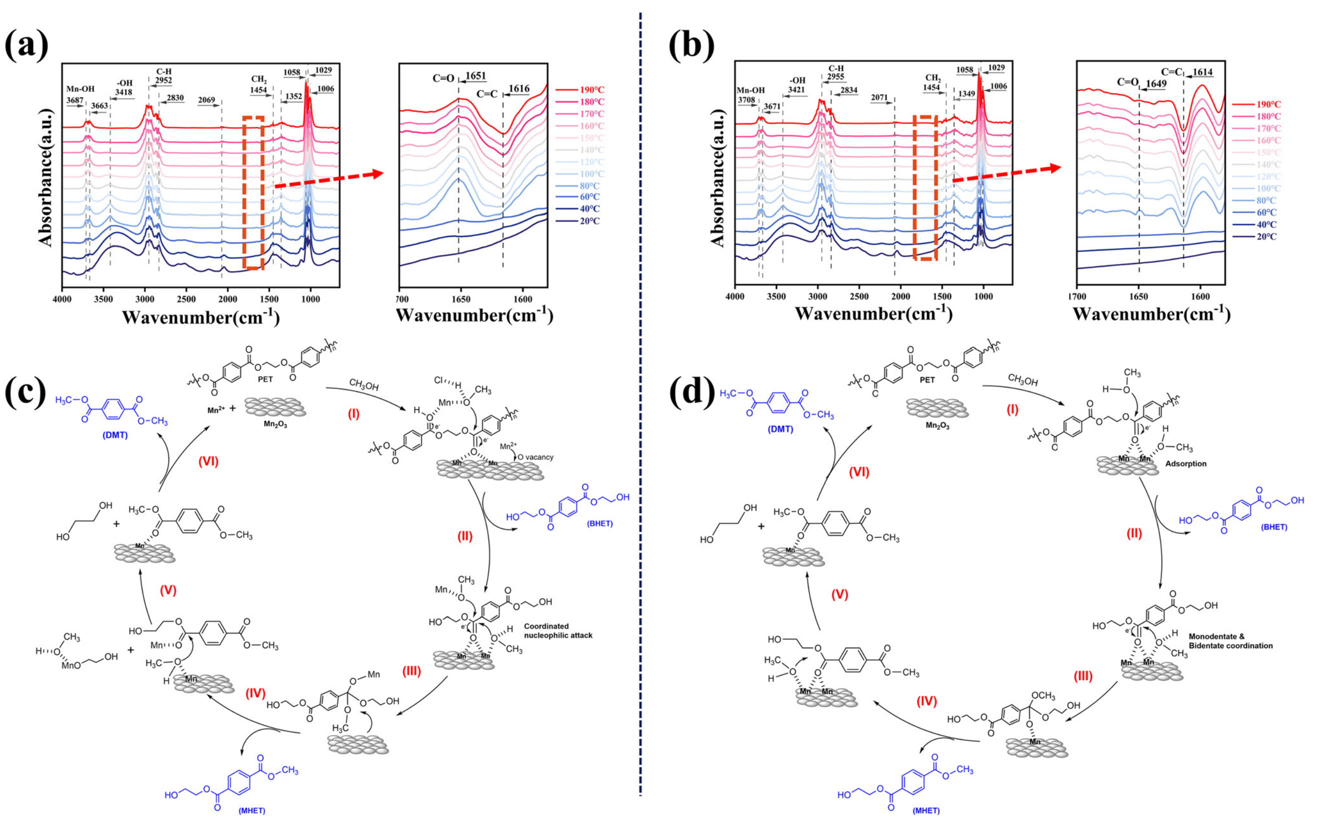

Fig. 4 *In situ* DRIFTS of (a) the synergistic system and (b) the heterogeneous system; mechanism of (c) the synergistic system and (d) the heterogeneous system.

likely accumulate at the interface of Mn_2O_3 particles *via* electrostatic interactions or weak coordination with surface oxygen vacancies[64,65] (whose increase was confirmed by post-reaction XPS). This enrichment modulates the local electronic structure at the interface,[47] enhancing the Lewis acidity of the Mn^{3+} sites on the Mn_2O_3 surface. Consequently, the electron cloud density of the adsorbed carbonyl group is more effectively lowered, making it more susceptible to nucleophilic attack. Simultaneously, homogeneous Mn^{2+} can rapidly coordinate with methanol in the initial reaction stage to form an active methoxymanganese intermediate $[Mn^{2+}–OCH_3]^+$. This intermediate acts as a strong nucleophile, directly attacking the ester carbonyl carbon at the chain ends of dissolved or swollen PET, thereby initiating the classic homogeneous transesterification pathway. In the infrared spectrum (Fig. 4a), the signal at ~1651 cm^{-1} (coordinated carbonyl) for the synergistic system is pronounced and diminishes rapidly as the reaction proceeds, with its intensity and decay rate far exceeding those in the Mn_2O_3-only system. This directly proves that the reactant (BHET) is more efficiently adsorbed and consumed at the synergistic interface. Furthermore, the synergistic system does not develop a strong, broad adsorption band in the characteristic carbonyl region of the product DMT (~1720 cm^{-1}), indicating that the generated DMT can promptly desorb from the active sites. This effectively prevents site poisoning and side reactions caused by strong product adsorption, which aligns perfectly with the experimental observations of higher product purity and lighter color for the synergistic system.

It is noteworthy that both systems exhibit a growing negative band around ~1616 cm^{-1}, corresponding to a key surface intermediate (potentially a manganese-stabilized enolate or a metal-alkoxy-carboxylate complex) that is being consumed. However, a significant difference exists in their band shapes: the absorption band in the synergistic system is noticeably broadened, whereas that in the Mn_2O_3-only system remains relatively sharp (Fig. 4a and b). This divergence carries important diagnostic significance. On the Mn_2O_3-only surface, the sharp band shape suggests that the intermediate resides in a relatively uniform and well-defined chemical environment, likely arising from its interaction solely with the fixed Mn^{3+} sites on the Mn_2O_3 surface. Conversely, in the synergistic system, the observed band broadening reflects the diversity of the intermediate's chemical environment. This spectral feature, together with the EPR results (Fig. 2c)—where fresh Mn_2O_3 exhibits an intrinsic oxygen vacancy signal at $g \approx 2.003$, and the synergistic system shows a significantly enhanced signal after the reaction while the non-synergistic system shows only a modest increase—and the increased O 1s peak intensity at ~531.7 eV after the synergistic reaction, collectively point to the formation of a dynamic $Mn^{2+}–O–Mn^{3+}$ composite interface. Within this region, the intermediate may bind to manganese sites of different electronic states (*e.g.*, Mn^{2+}, Mn^{3+}, or bridging sites) through varied microstructural linkages, resulting in a distribution of its vibrational frequencies that manifests as a broadened absorption band. This combination of *in situ* spectroscopic evidence with complementary *ex situ* characterization studies supports the conclusion that the synergistic effect arises not from a simple superposition of homogeneous and heterogeneous pathways but from the construction of a composite interface endowed with rich electronic structures and multiple active sites. Such an interface not only stabilizes the key reaction intermediate more effectively but also provides an optimized microenvironment for reactant activation, conversion, and product desorption.

In summary, *in situ* spectroscopic analysis demonstrates that the $Mn_2O_3/MnCl_2·4H_2O$ synergistic system optimizes the reaction pathway at the molecular scale by constructing a dynamic composite interface, thereby manifesting superior catalytic efficiency, selectivity, and product separability at the macroscopic level.

Aspen Process simulation and economic analysis

Based on the experimental results, an industrial-scale recovery process was further explored through Aspen Plus simulation. Direct water washing cannot efficiently remove $MnCl_2$ crystals encapsulated by organic products.[66] To address this, a water/cyclohexane biphasic system was employed. Based on the principle of dielectric constant-driven phase partitioning, cyclohexane ($\varepsilon \approx 2.02$) and water ($\varepsilon \approx 78.5$) exhibit a substantial polarity difference, ensuring that the ionic compound $MnCl_2$ preferentially partitions into the aqueous phase while DMT remains in the organic phase.[67] Subsequent phase separation yields an aqueous $MnCl_2$ solution for catalyst recovery by evaporation and crystallization, while the organic phase is directed to downstream separation steps to obtain high-purity product. Cyclohexane is fully recyclable with minimal loss, ensuring the environmental compatibility of the proposed flowsheet (see Fig. S28).

We simulated both the synergistic and the standalone Mn_2O_3 systems to quantitatively assess the material/energy consumption and economic performance of the $Mn_2O_3/MnCl_2·4H_2O$ synergistic catalytic process, in comparison with the non-synergistic process using Mn_2O_3 alone. To evaluate the economic feasibility of the proposed $Mn_2O_3/MnCl_2·4H_2O$ synergistic system relative to the non-synergistic one, an economic assessment was conducted using the built-in economic analysis module in Aspen Plus. Based on experimental results and under identical reaction conditions (180 °C, 1 h) and PET processing capacity (20 000 kg h^{-1}), the economic estimates for both processes were performed, with key results summarized in Tables S23–29.

The analysis reveals a significant economic advantage for the synergistic process. Compared with representative PET depolymerization systems reported in the literature,[68] the synergistic system also demonstrates competitive techno-economic performance, particularly in terms of catalyst cost and product purity. More importantly, a direct internal comparison with the non-synergistic system further highlights its inherent advantages.

Although it incurs a moderately higher total capital investment (USD 10.89 million *vs.* USD 8.30 million, +31.20%, as the cost of the separation operation unit calculated) and annual operating cost (USD 1415.79 million *vs.* USD 1300.53 million, +8.86%), these incremental costs are decisively offset by its

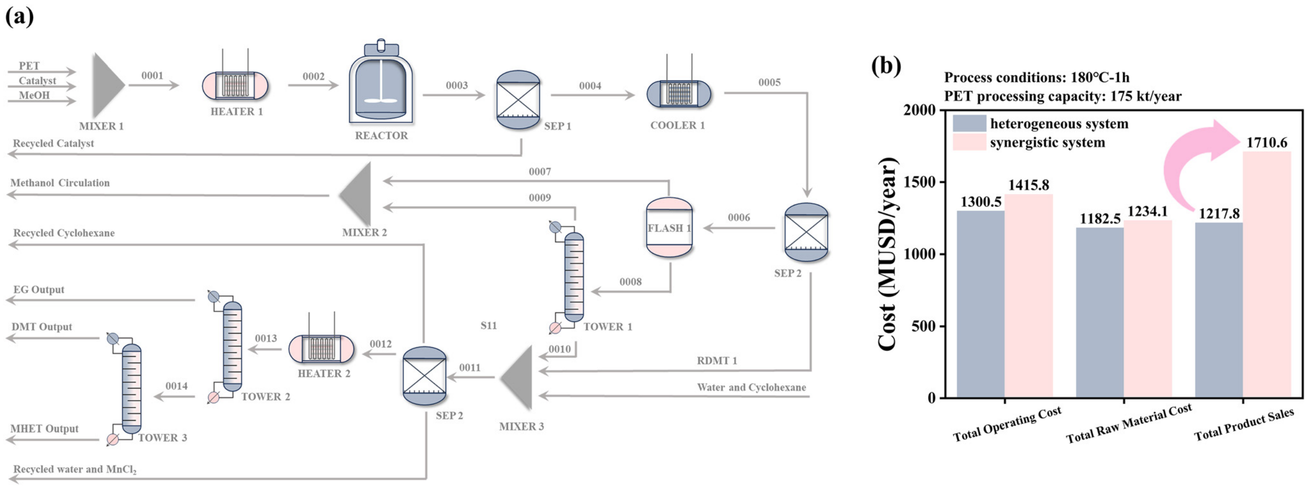

Fig. 5 (a) Aspen Plus process simulation of the synergistic system and economic evaluation; (b) comparison of economic effects between collaborative systems and heterogeneous systems.

superior revenue-generating capability. The synergistic system achieves a substantial 40.47% increase in annual product sale value (USD 1710.64 million *vs.* USD 1217.78 million), directly attributable to the experimentally demonstrated enhancements in DMT yield and selectivity. This fundamental disparity stems from the synergistic system's ability to convert raw materials into high-value products more efficiently, as evidenced by its significantly higher ratio of product sales value to raw material cost. To evaluate investment efficiency, the static payback period was employed as a key metric under the assumptions of a 10-year straight-line depreciation for equipment and a corporate income tax rate of 25%. Owing to its considerable EBITDA, approximately $29.48 million per year, the synergistic system yields an estimated annual net profit after tax of about $18.86 million after accounting for depreciation and taxes, which results in a calculated static payback period of approximately 5.8 years. In contrast, due to its lower product yield, the annual operating cost of the non-synergistic system already exceeds its sales revenue, placing it in an operational loss state with a negative EBITDA, which makes its payback period theoretically approach infinity, implying that this process route cannot achieve capital recovery under the given economic assumptions and lacks financial feasibility.

Therefore, despite introducing an additional separation step for homogeneous catalyst recovery, the Mn_2O_3/$MnCl_2·4H_2O$ synergistic catalytic process establishes a clear economic superiority by translating catalytic performance gains—namely, lower activation energy and higher selectivity—into decisive financial metrics. This validates its potential for commercial application (Fig. 5).

Conclusion

In this study, we have developed a synergistic catalytic system comprising homogeneous $MnCl_2·4H_2O$ and heterogeneous Mn_2O_3 for the efficient methanolysis of waste polyethylene terephthalate (PET). This system achieved complete PET conversion with a dimethyl terephthalate (DMT) yield of 95.35% within 1 hour at 180 °C. Kinetic analysis indicated that the synergistic effect reduced the apparent activation energy to 151.61 kJ mol^{-1}. *In situ* characterization studies revealed that the performance enhancement originated from a dynamic composite active center formed at the interface between Mn^{2+} and Mn_2O_3. This interface not only efficiently stabilized key reaction intermediates but also facilitated product desorption. The catalysts could be recovered in a straightforward manner: the heterogeneous Mn_2O_3 was directly reusable *via* centrifugation, while the homogeneous $MnCl_2$ was effectively removed by water washing, providing a basis for industrial recovery *via* extraction, evaporation, and crystallization. Process simulation and economic analysis confirmed that the synergistic system, by virtue of its higher product yield, could deliver an approximately 40.5% increase in product sale value, with a payback period for the incremental investment of only about 5.8 years, demonstrating clear economic advantages for industrialization compared with the non-synergistic system. In summary, this work provides a green, efficient, and economically viable solution for PET chemical recycling by constructing a novel synergistic catalytic interface using low-cost, commercially available manganese sources.

Author contributions

Lei Yang: conceptualization, data curation, formal analysis, investigation, methodology, validation, visualization, software, writing – original draft, and writing – review & editing. Jiacheng Gao: conceptualization, methodology, investigation, supervision, and writing – review & editing. Li Yuan: resources, validation and visualization. Like Ouyang: conceptualization, methodology, project administration, supervision, funding acquisition, and writing – review & editing.

Conflicts of interest

There are no conflicts to declare.

Data availability

The data supporting this article have been included as part of the supplementary information (SI). Supplementary information is available. See DOI: https://doi.org/10.1039/d6gc01939k.

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